

Unit 5, Lesson 7: Using the Pythagorean Theorem to Solve Real-World Problems

Benchmark

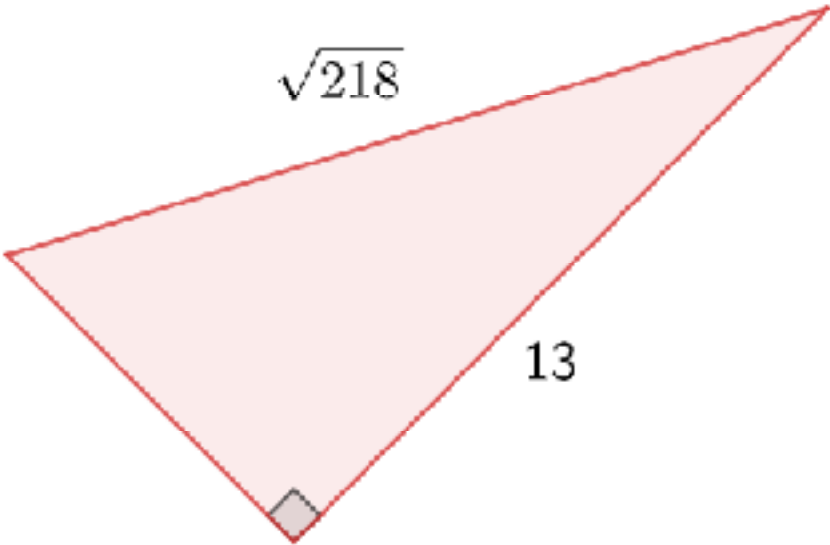
MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

Learning Target

- ☐ I can solve real-world problems by applying the Pythagorean Theorem.

5.7.1 Warm-Up: Bell Work

Sawyer applied the Pythagorean Theorem to find the length of the missing side of the triangle. His work is shown.



$$\begin{aligned} a^2 + b^2 &= c^2 \\ 13^2 + \sqrt{218}^2 &= c^2 \\ 169 + 218 &= c^2 \\ 387 &= c^2 \\ \sqrt{387} &= c \end{aligned}$$

The missing side is $\sqrt{387}$ units long.

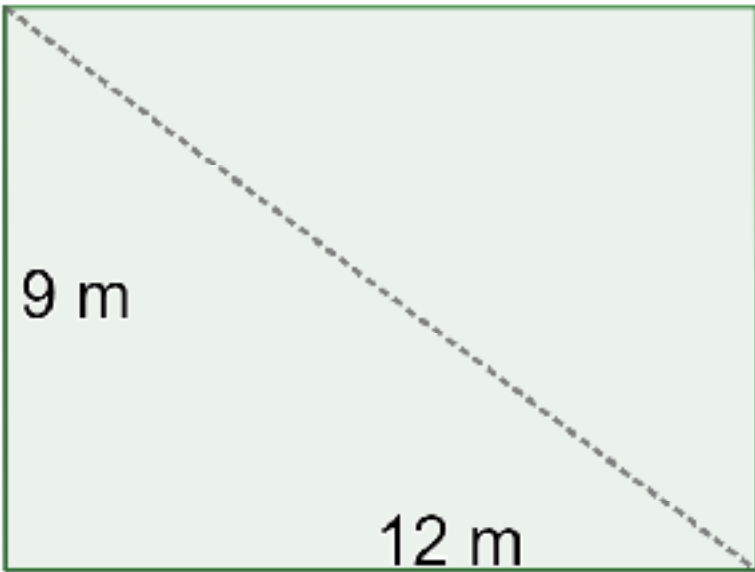
- 1. Write a note to Sawyer explaining his mistake.
- 2. Correct Sawyer's mistake.

5.7.2 Collaboration: Who Wins?

Mai and Tyler were standing at one corner of a large rectangular field and decided to race to the opposite corner. Since Mai had a bike and Tyler did not, they thought it would be a more equal race if Mai rode along the sidewalk that surrounds the field while Tyler ran the shorter distance directly across the field.

The field is 12 meters (m) long and 9 m wide.

- 1. Tyler can run at around 5 m per second, and Mai can ride her bike at around 7.5 m per second.



- a. Before making any calculations, predict who you think will win and by how much. Show or explain your thinking.
- b. Work with your partner to mathematically determine who will win the race.

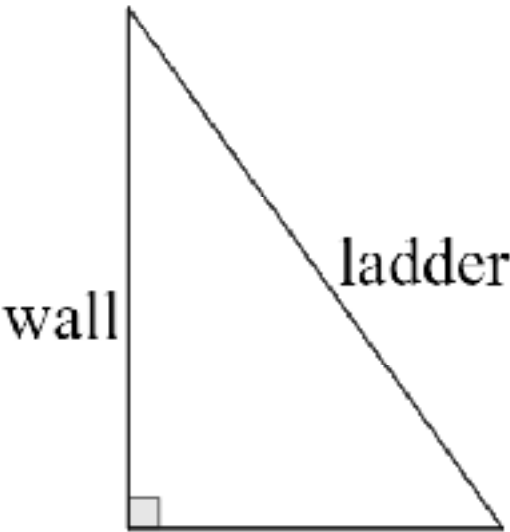
5.7.3 Guided Instruction: Applying the Pythagorean Theorem

The Pythagorean Theorem can be used to find an unknown side length of any right triangle.

Pythagorean Theorem Problem-Solving Steps

- 1. Model the problem with a diagram (if not provided).
- 2. State the Pythagorean Theorem.
- 3. Substitute given values.
- 4. Solve for the unknown.

- 1. A 12-foot long ladder leans against a vertical wall. The base of the ladder is 7 feet from the bottom of the wall. Determine, to the nearest tenth of a foot, how high the ladder reaches up the wall.



- a. Label the diagram with the given information.
- b. State Pythagorean Theorem
- c. Substitute given values
- d. Solve for the unknown

$$a^2 + b^2 = c^2$$
$$\underline{\hspace{1cm}}^2 + \underline{\hspace{1cm}}^2 = \underline{\hspace{1cm}}^2$$

- 2. A cable is being placed on level ground to support a tower. The cable is 17 feet (ft) and should be connected 15 ft up the tower. How far away from the bottom of the tower should the other end of the cable connect to the ground?

- a. Model the problem.

- b. State Pythagorean Theorem

$$a^2 + b^2 = c^2$$

- c. Substitute given values

$$\underline{\hspace{1cm}}^2 + \underline{\hspace{1cm}}^2 = \underline{\hspace{1cm}}^2$$

- d. Solve for the unknown.

- 3. A soccer field is 115 yards (yd) long by 70 yd wide. What is the length of the diagonal of a soccer field? Round your answer to the nearest tenth.

- a. Model the problem.

- b. State Pythagorean Theorem

$$a^2 + b^2 = c^2$$

- c. Substitute given values

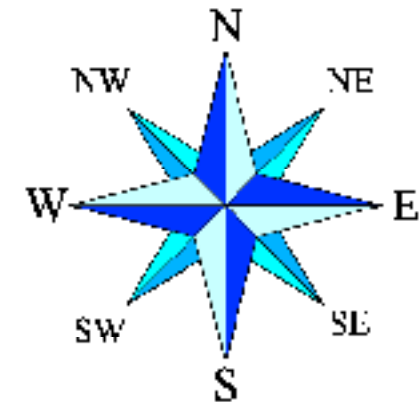
$$\underline{\hspace{1cm}}^2 + \underline{\hspace{1cm}}^2 = \underline{\hspace{1cm}}^2$$

- d. Solve for the unknown

5.7.4 Guided Practice: Applying the Pythagorean Theorem

Solve each problem using the Pythagorean Theorem.

1. Malik leaves his house and drives 7 miles (mi) west. He then turns and drives 5 mi north. How far away is he from his house? Round your answer to the nearest tenth.

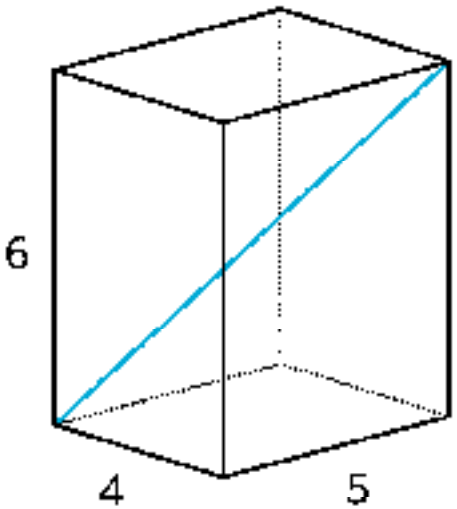


2. Audrey designs a 15-foot slide from her bedroom window into her swimming pool. The end of the slide is 9 feet from the base of her house. How high up is her window? Round your answer to the nearest tenth.

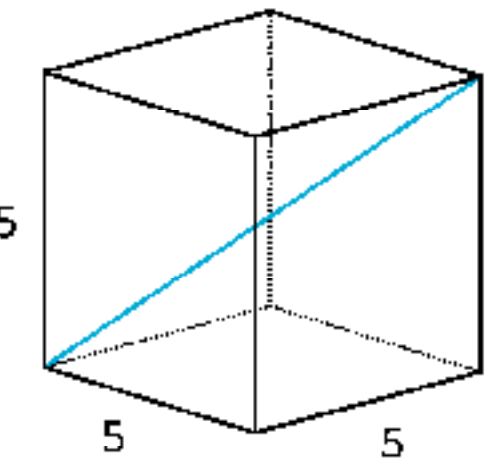
5.7.5 Collaboration: Internal Dimensions

1. Two rectangular prisms are shown:

K



L



- a. Predict which figure has a longer diagonal. Note that the figures are not drawn to scale.
- b. Discuss with your partner if there is enough information to find the diagonal using the Pythagorean Theorem. Summarize your discussion.

- c. What other measurement must be calculated first to find the diagonal? Label this extra measurement on the figures above.

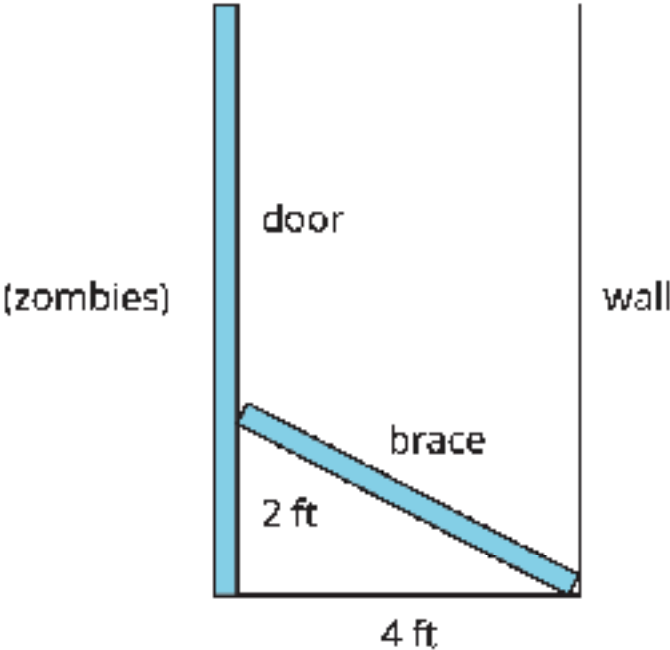
- d. Find the exact length of the base diagonals using the Pythagorean Theorem.

- e. Use the Pythagorean Theorem to calculate the length of both diagonals. Which one is longer?

5.7.6 Your Turn!

Solve each problem by applying the Pythagorean Theorem.

- 1. A man is trying to zombie-proof his house. He wants to cut a length of wood that will brace a door against a wall. The wall is 4 feet (ft) away from the door, and he wants the brace to rest 2 ft up the door. About how long should he cut the brace?



- 2. Lovelace Park in Jacksonville is a rectangular-shaped park, measuring approximately 410 meters (m) by 360 m. Joseiré wants to get from one corner of the park to the opposite corner. She wonders about the difference between cutting across the diagonal through the park compared to going around the park, along the streets.

How much shorter would her walk be cutting through the park? Round your answer to the nearest tenth.

3. At a restaurant, a trash can’s opening is rectangular and measures 7 inches (in.) by 9 in. The restaurant serves food on trays that measure 12 in. by 16 in. Jada says it is impossible for the tray to accidentally fall through the trash can opening because the shortest side of the tray is longer than either edge of the opening.

Do you agree or disagree with Jada’s explanation? Justify mathematically.

5.7.7 Wrap-Up: Ticket Out the Door

1. Jonathan places a 14-foot ladder against the side of the school. The top of the ladder is 11 feet above the ground.

How far from the building is the base of the ladder? Round your answer to the nearest tenth.

